

robolabel: Video-First Labeling of Robot and Human Demonstrations with Vision-Language Models

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Abstract

Policies that condition on more than a one-line instruction need structured labels for their demonstrations: the phases of the task with their frame boundaries, each grasp attempt and whether it failed, and the goal. We present robolabel, an open-source tool that drafts these labels from video with vision-language models (VLMs) and measures how far the drafts can be trusted. We describe three generations of its pipeline and the measurements behind each change. A contact-sheet design capped boundary recall within ± 5 frames at 0.58 on the SO-101 development episodes, by construction. A second design that anchored boundaries to the robot’s gripper signal raised one rater’s mean score from 1.5 to about 3 of 5, but made nine models nearly interchangeable: they confirmed 252 of the 270 signal candidates they were shown. The current design, v1.1, works from video alone and treats robot signals as optional hints. A coarse pass proposes typed phase boundaries from frames sampled at 2 per second, and a crawl refines each finger or contact boundary with two narrowly posed questions over dense eight-frame windows. We test boundary timing on 60 development episodes from two SO-101 datasets, with the gripper signal hidden from the models and used only as truth. Against the measured gripper onset, the crawl raises typed boundary F1 at 5 frames by 0.077 (95% CI 0.036 to 0.119), with the whole gain on one of the two datasets; against the commanded onset, 3 to 4 frames earlier, it lowers F1 by 0.060 (-0.102 to -0.017). In every configuration, 64.5 to 73.0% of true gripper events have no prediction of the same type within 1 s. Gemini’s native video input scored below sampled frames against the measured onset, and motion-derived hints made no clear difference. Across six models on six domains (a robot arm, a humanoid, egocentric and third-person human video, a simulated drawing hand and a dance), every run returned schema-valid labels and the median cost per clip ranged from \$0.006 to \$0.522. One rater’s mean scores ranged from 2.17 to 3.17 of 5, and every model produced unusable labels on at least two of the six clips. The v1.1 experiments cost \$8.16 in API fees. The code and prompts are open source.

1 Introduction

Robot learning datasets ship video, proprioception and, at best, one line of text per episode [1, 2, 3]. The policies trained on them increasingly want more. Hierarchical and chain-of-thought policies condition on the current subtask [4, 5, 6, 7]; curation needs to know which attempts failed and why; goal-conditioned methods need the end state an episode was meant to reach. Producing that structure by hand is slow, and the public datasets that would benefit most, community collections in the LeRobot format [8], are the least consistently annotated [9]. Vision-language models (VLMs) can draft these labels cheaply. The difficulty is that a fluent draft can be quietly wrong, and a labeling tool that hides that failure mode does more harm than good.

robolabel started from that observation. In a first review of 100 BridgeData V2 episodes labeled by a VLM, a human reviewer changed 58 of the 100 quality scores and matched the model’s subtask boundaries at

a temporal IoU of only 0.683. The project’s stated aim since then has been labels plus a measurement of how wrong they are, plus a loop for fixing them. This paper reports the current state of that effort in full, including the designs that did not work.

The labels robolabel drafts for one episode are: the phases of the activity, each with a frame span, a short open-text description, a coarse class and a typed end event (the fingers start to close, start to open, a contact starts or ends); the attempts at each step and the outcome of each phase and each attempt, with the failure marked on the phase that failed; the goal as a list of end-state requirements over objects and the robot, together with a command rendered from those requirements; whether the activity has an object end state at all; and scene facts and checks that flag episodes for review. All of it is written to a long-format parquet file, one row per record, with a receipt for every model call. Figure 1 shows four examples.

Getting the boundaries right has been the hardest part, and it is where most of this paper’s measurements sit. Three findings shaped the design.

Sparse frames cap timing. The first generation (Section 3.1) showed the model 12 evenly spaced frames on one contact sheet and asked for all the segment fields in one labeling pass. Every boundary it returned sat exactly on one of the 12 sampled frames, so on the SO-101 development set no model could place more than 58% of the human boundaries within ± 5 frames. A reported gain in boundary recall from a stronger strategy on a stronger model (Flash S0 to Pro S2, 0.226 to 0.307 on 20 held-out episodes) turned out to be inconclusive under a paired bootstrap.

Signals anchor timing and hide the model. The second generation, V-lite (Section 3.2), split the job into four small calls and took timing from the robot’s gripper signal. One rater’s mean score rose from 1.5 for the old prompts to a median of 3.0 across nine models. But the signal did most of the work. The models confirmed 252 of the 270 signal candidates they were shown, their boundaries at the confirmed candidates were identical by construction, and a signal-only arm with no model at all scored at the median model. The largest remaining errors came from resets after missed grasps, which the signal’s candidate list did not cover.

Video first, with refinement by rule. The third generation, v1.1 (Section 4), is built on the principle that every layer must work from video alone, with robot signals as optional hints. A coarse pass reads frames at 2 per second (at most 48) and proposes typed boundaries. A crawl then revisits every boundary typed as a finger or contact event: it shows the model 8 frames spanning ± 1 s around the proposal and asks which is the first to show the event, then repeats at native or near-native spacing between the chosen frame and the one before it. The crawl runs on every such boundary by rule, since models rarely flag their own uncertainty.

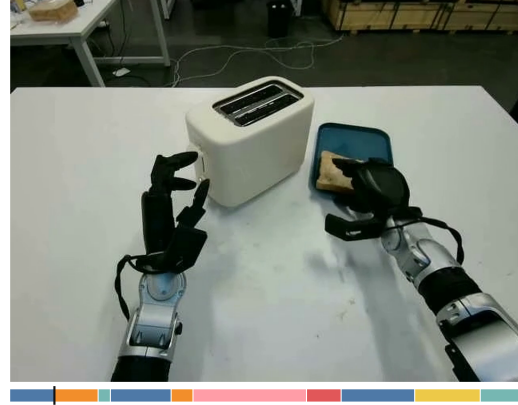
We evaluate v1.1 in three experiments (Sections 5 and 6). E1 measures boundary timing on 60 development episodes from two SO-101 datasets, with the gripper signal hidden from every model call and used only to score. Against the measured gripper position, the crawl gives a clear improvement in typed boundary F1 at 5 frames, +0.077 (95% CI 0.036 to 0.119), all of it on one of the two datasets. Against the commanded gripper position, which moves 3 to 4 frames earlier, the same crawl makes timing worse, -0.060 (-0.102 to -0.017). The refined onsets land a median of 4 frames after the measured onset, which suggests the crawl finds the moment the motion becomes visible; which of the two onsets a label should mark is a choice the data do not settle. Gemini’s native video input, as the provider samples it, scored below frames at 2 per second against the measured onset, and motion-derived hints made no clear difference. The dominant error is omission: in every configuration, 64.5 to 73.0% of true gripper events have no prediction of the same type within 1 s. E2 runs the full pipeline with six models on six clips from six domains: every output was schema-valid, the median cost per clip spanned a factor of 84, and every model produced labels that one rater judged unusable on at least two of the six clips. E3 finds that no model forced a dance into pick-and-place phases or invented an end state for it.



Gripper attempts to grasp the eye drops box but slips and misses

Goal: Put the eye drops box in the basket
All fields from raw video in 2 min 20 s
Total cost: \$0.164

(a) SO-101 arm (Gemini 3.8 Flash; 2 min 20 s, \$0.164).



Right hand tries to grasp bread but fails to lift it

Goal: Put the slice of bread in the white toaster then press the toaster lever
All fields from raw video in 1 min 22 s
Total cost: \$0.263

(b) Humanoid (Claude Opus 5.5; 1 min 22 s, \$0.263).



Opens the T-shirt out and lays it back on the tabletop.

Goal: The white "GOOD VIBES ONLY" T-shirt is folded on the tabletop.
All fields from raw video in 35 s
Total cost: \$0.060

(c) Egocentric (GPT-6 Sol; 35 s, \$0.060).



Scrub the inside and outside of the bowl with the sponge, turning it under the water.

Goal: Set the white plate under the bowl to clean
All fields from raw video in 1 min 28 s
Total cost: \$0.121

(d) Third person (GPT-6 Sol; 1 min 28 s, \$0.121).

Figure 1: Labels that robolabel v1.1 drafted from raw video in four domains, as frames of the animated figures in the project README. Each shows the clip, the labeled timeline, the current phase (outline and text in the phase color, light red fill for a failed phase), the rendered command or objective, the wall time of the whole pipeline and its API cost. Three panels are cards the rater found impressive and asked to turn into figures; the humanoid panel is the rater’s top-rated card on that clip, which replaced the requested card after the ratings were saved. They are not typical (Section 6.5). Clips: (a) ArmNetBench v0.1, Apache-2.0; (b) Unitree G1_Dex3_ToastedBread_Dataset, Apache-2.0; (c) Eidon Tracker POV recording 8266 by Eidon AI, CC BY 4.0; (d) “Washing Dishes” by Leet289, Wikimedia Commons, CC BY-SA 4.0, so panel (d) is shared under CC BY-SA 4.0.

Contributions.

- An open-source pipeline that drafts phases with typed boundaries, attempts with per-phase failure marks, goals as end states with a rendered command, and a flag for activities without an end state, from one camera’s video, with robot signals as optional event sources (Section 4).
- The crawl, a coarse-to-fine boundary refinement that asks narrow questions over dense frame windows and runs on every finger or contact boundary by rule (Section 4.5).
- A free timing evaluation that uses hidden robot signals as truth, scores typed boundaries against two plausible definitions of onset, and reports paired cluster bootstrap intervals with pre-declared minimum effects and three fixed outcome words (Sections 5 and 6.1).
- A comparison of six models on six domains covering cost, speed, schema validity, accuracy on the clips that have truth, the generality of the labels, and one rater’s ratings in two rounds (Sections 6.2 to 6.4).
- An account of three generations of the design, with the negative and null results reported alongside the positive ones (Section 3), and the infrastructure that made small-budget evaluation trustworthy: a spend guard that reserves every request’s worst-case cost, a content-addressed response cache, per-call receipts, and a guard that seals held-out episodes (Section 4.9).

2 Related Work

Language and structure in robot demonstration data. Large robot datasets pair trajectories with a single free-form instruction per episode [1, 2, 3]. Policies that condition on more than that instruction need more structure in the data: RT-H inserts intermediate “language motions” between the task and the action [4], $\pi_{0.5}$ trains on high-level subtask annotations alongside low-level actions [5], Hi Robot relabels demonstrations with hierarchical instructions [6], and embodied chain-of-thought policies are trained on automatically generated plans, subtasks and object locations [7]. Community datasets in the LeRobot format [8] are the most accessible source of such data and also the least consistently annotated; SmolVLA, trained on those datasets, regenerated their task descriptions with a vision-language model before training [9]. robolabel targets exactly this gap: per-phase subtask text with frame boundaries, grasp attempts and their outcomes, and goals, for datasets that ship with one instruction or none.

Automatic labeling with foundation models. DIAL relabels robot trajectories with instructions proposed by a fine-tuned CLIP model [10]. NILS segments long-horizon robot play into subtasks at detected keystates and labels each segment with language, zero-shot, by combining several pretrained models [11]. Vision-language models have been used as success detectors for robot episodes [12] and as annotators for embodied reasoning traces [7]. UVD finds subgoals as changes in a pretrained visual representation [13]. robolabel differs in three ways: it produces a fixed, checkable schema per episode (phases, attempts, goals, keyframes) with a record of every model call and its cost; it treats the robot’s own signals as optional evidence rather than a requirement, so the same pipeline runs on humanoid and human video; and it measures its boundary timing against signals the models never see.

Temporal segmentation and boundary detection. Temporal action segmentation assigns a label to every frame of a video [14, 15]; generic event boundary detection marks the instants where one event gives way to another and scores predictions with an F1 at a relative distance tolerance [16]. We borrow the boundary view and its tolerance-based F1, with absolute frame tolerances because a grasp onset has a physical meaning, and

Table 1: Three generations of robolabel. “Frames seen” is what the model sees when it places boundaries.

	v6 (June 2026)	V-lite (September 27, 2026)	v1.1 (September 28, 2026)
Frames seen	12 evenly spaced frames on one contact sheet	about 20 evenly spaced frames plus frames around each gripper event	frames at 2 per second (at most 48), then dense 8-frame windows per boundary
Who sets timing	the model, on the 12-frame grid	the gripper signal; model boundaries snap to it	the model from video, refined by the crawl; a signal is optional
Model calls	observe and label (2 to 7 per episode)	inventory, scene facts, segments, goal	inventory, coarse pass, crawl, scene facts, goal
Phase names	closed pick-and-place vocabulary, later free text	16 classes plus text	open text plus a class, typed end events
Failures	a per-episode flag	every phase of a failed attempt marked failed	only the phase that failed; attempt outcome kept separately
Goal	the task string	end-state requirements	end-state requirements, a rendered command, and <code>has_end_state</code>
Without robot signals	yes	no	yes (the default)
Measurement	legacy gold anchored to model drafts	harness, blind ratings, receipts	plus timing against hidden signals

we add the matched-pair mean absolute error and a miss rate. Vision-language models are known to be weak at precise temporal localization when they see sparse frames [17, 18]; iterative frame selection improves long-video question answering [19]. The crawl in Section 4.5 applies the same idea to a single boundary: a coarse proposal followed by two dense, narrowly posed questions.

Human and humanoid video. Egocentric and exocentric human video is a growing source of manipulation data [20, 21, 22], and humanoid platforms with dexterous hands now release LeRobot-format datasets of their own [23]. Labels for such video cannot come from a gripper channel. A pipeline that works from the camera alone, and uses a signal only when one exists, is the design constraint that shapes robolabel v1.1.

Evaluation methodology. Our statistics follow standard practice for small, paired, clustered samples: a paired cluster bootstrap over episodes [24, 25] with pre-declared minimum effects of interest and equivalence-style outcome words [26]. Human ratings are collected on a blind page, with outputs lettered in a random order per clip, the same precaution that public model comparisons take when they hide model identities until a vote is cast [27]; Section 6.4 reports where this blinding failed.

3 Three Generations of the Pipeline

robolabel has been rebuilt twice. Each rebuild followed a measurement that exposed a limit of the previous design. We summarize the two earlier generations here because their negative results motivate v1.1, and because they are not reported elsewhere. Table 1 compares the three.

Table 2: The frame-grid ceiling on the SO-101 development episodes (30 episodes, 183 to 303 frames, 86 human boundaries): the share of human boundaries that lie within τ frames of at least one sampled frame, when a model can only answer with sampled frames.

Frames sampled	Mean spacing (frames)	$\tau = 3$	$\tau = 5$	$\tau = 10$
6 (S0)	48.7	0.116	0.151	0.384
12 (S1 to S4)	22.1	0.360	0.581	0.942
24	10.6	0.674	0.988	1.000
32	7.8	0.849	1.000	1.000
48	5.2	1.000	1.000	1.000

3.1 v6: one contact sheet, one question

The first public version (schema `robolabel/annotations/v6`) was built over one week in June 2026 (June 10 to 16). Per episode it drafted subtask boundaries with a phase and a target, an episode quality score from 1 to 5, and subgoal keyframes (the last frame of each segment), plus deterministic speed, control-mode and curation fields computed without a model. All sampled frames were tiled into a single JPEG contact sheet, which Gemini read as one image of 258 tokens, about 21 tokens per frame. A two-stage prompt first asked the model to list the physical events it saw with frame indices and then to segment the episode.

The strategy ablation. With Gemini 2.5 Flash and Gemini 2.5 Pro, we compared five annotation strategies against a 50-episode human gold set on the PickPlace dataset (`lerobot/svla_sol101_pickplace`): S0, a plain prompt on 6 frames; S1, 12 frames captioned with their index and time and a boundary grounded in a cited frame with a one-clause evidence string; S2, S1 plus a closed phase vocabulary and a minimum of three segments; S3, S2 plus a dense-window refinement of each boundary; and S4, S3 plus three samples with a per-boundary median. Selection used a frozen 30-episode tune split and a single, once-only evaluation on 20 held-out episodes. The winning cell on tune, Pro with S2, beat the Flash S0 baseline on mean boundary IoU (0.453 against 0.400) by a margin of 0.053, three thousandths above the pre-declared 0.05 bar. On the held-out episodes it lost (0.444 against 0.460). Fewer outputs were catastrophic with grounding: the share of held-out episodes that came back as a single “complete the task” segment or as five uniform segments fell from 5 of 20 (Flash S0) to 0 of 20 (Pro S2), and on a fresh stacking dataset that no one had labeled, from 8 of 20 to 0 of 20 for the same model. Pro with S2 also placed more gold boundaries within ± 5 frames on the held-out episodes, 19 of 62 against 14 of 62. We later re-scored that difference with a paired cluster bootstrap over episodes: $+0.08$, 95% CI -0.05 to $+0.21$, inconclusive. The comparison also changed the model and the strategy at once.

Why v6 hit a ceiling. A zero-cost audit of the cached June receipts found two structural limits. First, every boundary the grounded strategies returned on the 30 development episodes (123 for Flash S1, 121 for Flash S2, 120 for Pro S2) sat exactly on one of the 12 captioned frames: the prompt asked for an end frame “from the captions above” and the models complied. With 12 evenly spaced frames, the best possible recall of the human boundaries is 0.36 within ± 3 frames, 0.58 within ± 5 and 0.94 within ± 10 (Table 2). The sampling capped timing precision. Second, the human gold could not serve as a clean reference. The review tool could change the boundaries and text of drafted segments but could not add or delete one, so every gold episode had exactly as many segments as its S0 draft; 5 of the 30 development golds were the parse-failure fallback “complete the task”; and 106 of 116 gold subtask texts were identical to the draft.

Two refinement attempts inside v6 did not help. A dense-window refinement restricted to contact boundaries left recall within ± 5 frames unchanged at 0.211 on 8 episodes, and a deterministic snap of grasp and release boundaries to the nearest gripper transition lowered the recall of those boundaries from 0.174

to 0.130 (4 of 23 to 3 of 23), because the human gold itself was anchored to the S0 draft rather than to the gripper. Separately, an unpublished controlled test in the author’s world-model testbed [33] found no benefit from these conditioning labels as world-model training inputs, so the project was scoped to policy conditioning and dataset curation.

3.2 V-lite: small calls anchored to the robot’s signals

V-lite (schema v7, built in one night on September 26 to 27, 2026) replaced the contact-sheet calls with five layers. L1 reads the gripper channel deterministically: it finds closing and opening runs with a velocity threshold, groups them into grasp attempts, labels each attempt’s outcome (held, empty, slipped, released, aborted) from the gap between the commanded and measured gripper positions, and proposes candidate boundaries at attempt closings, release openings and the moments the arm starts to move after them. L2 lists the objects in the scene with identifiers and records scene facts at keyframes chosen around the gripper events. L3 segments the episode from about 20 evenly spaced frames plus frames 0.5 s before and after each gripper event, with the instruction to keep the signal’s candidates as boundaries unless the images clearly contradict them. L4 writes the goal as a list of end-state requirements, each marked required, incidental or unsure. L5 runs ten rules (for example, every missed grasp found by the signal must be matched by a failed segment) and routes risky episodes to review.

The model sweep. We ran V-lite with nine models through OpenRouter on four development episodes from three datasets: episode 0 of the PickPlace dataset of Section 5.1 (an SO-101 pick and place), two SO-101 policy rollouts from ArmNetBench, episode 544 (a failed block stacking with 11 grasp attempts) and episode 1821 (eye drops put in a basket after failed grasps), and a LIBERO simulation episode that picks one of two identical black bowls [28]. The whole sweep cost \$3.49. Every model accepted strict JSON-schema output. Two models, DeepSeek V4.1 Flash and MiMo v2.6 Pro, spent up to about 69,000 hidden reasoning tokens per episode at a “low” reasoning effort, took 28 to 109 minutes per episode and left several calls empty; Qwen3.8 Max returned two invalid scene-fact calls. The cheapest model, GPT-6 Luna, cost \$0.0036 per episode and the most expensive, GPT-6 Astra, \$0.34.

What one rater saw. The author rated all 42 outputs in quick mode (an overall score from 1 to 5 and whether the output is usable as a training label), with the cards lettered in a random order per episode. The letters came from a seed derived from the run and the episode rather than from a secret salt, and one card may have been exposed by a report table before rating. The old v6 prompts, run on Gemini 3.8 Flash as a baseline, averaged 1.50 and came last or tied for last on 3 of the 4 episodes. The nine V-lite models averaged 2.00 to 3.25, median 3.00. A signal-only arm that used no model at all averaged 3.00, the median. Only one episode, the clean pick and place, received any “usable as is” ratings. On the pick-and-place episode, every V-lite model that answered placed its boundaries at the same four signal candidates (frames 160, 176, 215 and 232), and three also added one at frame 95, so the automatic timing metrics said little about boundary skill. Across all V-lite outputs the models confirmed 252 of the 270 candidates they were shown and rejected 18, all on one episode. The boundary metrics were measuring the signal layer.

The review that became v1.1. The author’s notes after rating, as recorded at the time (paraphrased), and the diagnosis of each, were:

1. *Many models return the same outputs and the same boundary frames.* The signal fixed the structure, and the subtask texts were rendered from one template.
2. *Goals read as states (“the brick is inside the box”) where a command (“put the brick in the box”) is wanted.* The state form is checkable and useful for planners; training prompts need the command.

3. *An approach is marked failed when the grasp after it fails.* This followed our own convention, which marked every phase of a failed attempt as failed. For training labels the approach usually succeeded.
4. *Resets, retracts and re-approaches after a missed grasp cause most errors.* The signal’s candidate list had no event for the re-opening or the back-off, and on ArmNetBench episode 1821 five models folded each reset and re-approach into one long failed grasp. The two models that split those spans with approach segments of their own were rated 4, Luna, with retract segments only, 3, and the other five 2.
5. *The stages feel hard-coded; they would not cover a person picking up a paintbrush, or dancing.* The phase list and the gripper signal were shaped around pick and place.

A side discussion added two directions: signals should improve results when present but never be required, and boundaries need more frames plus a per-boundary refinement triggered by rule. Each note maps to a v1.1 change: a main arm with no signal, so each model sets its own boundaries (note 1); `goal_command` beside a state objective (note 2); the per-phase failure convention with `attempt_outcome` (note 3); recovery events and a recovery convention (note 4); and open-text phases, `has_end_state`, and test clips from humanoid, human and non-manipulation domains (note 5).

4 The v1.1 Pipeline

4.1 Overview

The design rule of v1.1 is that every layer must work from the video of one camera: robot data without gripper channels, web video, humanoids, and egocentric or exocentric human video, including activities that are not pick and place. A robot signal is an optional *event source*. When one exists it can improve timing, attempts and the robot’s end state; when it is absent nothing breaks, and a run records which source it used. Two further rules follow from the V-lite results. Candidate events from any source are hints, never boundaries to copy. And refinement runs by rule on every finger or contact boundary, never only when the model reports that it is unsure: in the author’s earlier work on state descriptions for language-model planners, models rarely flagged their own uncertainty [34].

Figure 2 shows the steps. The event source runs first and costs nothing. The scene inventory lists the objects that matter, with identifiers, from up to 8 evenly spaced frames. The coarse pass proposes every phase of the clip with a typed end event. The crawl refines each typed boundary to its onset frame. Scene facts are then read at keyframes placed at the refined boundaries. The goal call writes end-state requirements and whether the activity has an object end state, and a deterministic renderer turns the required object end states into a command. Finally, 13 rules check the result and route risky episodes to review. A failed, invalid or refused call does not end the episode: the gap is recorded, the episode’s risk is set to 1.0 and it is routed with the reason. Every model image is downscaled to a long side of at most 448 pixels and JPEG-encoded at quality 85, and each is preceded by a caption such as “frame 160 of 303 (5.33 s), camera up”.

4.2 Event sources

An event source returns a sorted list of typed candidate events, each with a frame, a confidence, a source label and an optional attempt index. The event types are `close_start`, `open_start`, `arm_move`, `pause_start`, `pause_end` and, for recovery, `back_off`. The coarse prompt shows them as plain lines (“c1: frame 152 (5.07 s), pause_start (motion)”) under the instruction “The candidates are hints, not boundaries to copy. Put a boundary only where you see an event, at a candidate or anywhere else, and leave out a candidate that you do not see happen.”

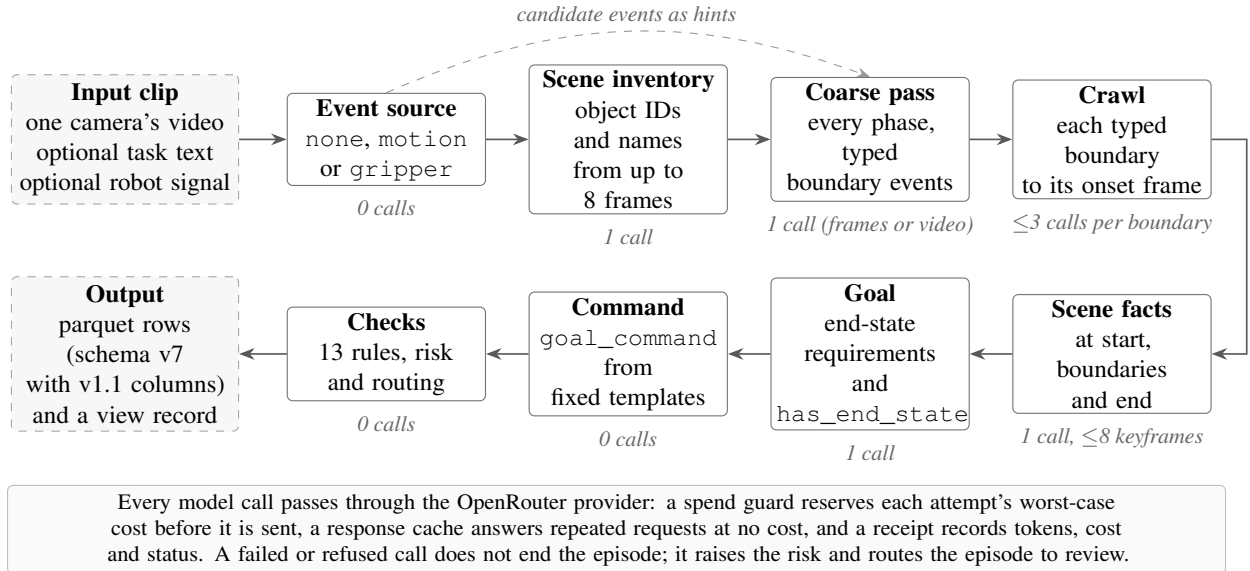


Figure 2: The v1.1 pipeline, `robolabel.vfirst.run_episode_v11`. A full episode makes between 3 and 40 structured model calls: one inventory call, one coarse call, up to 36 crawl calls (at most 3 per boundary and 12 boundaries), one scene-facts call (skipped when the inventory is empty) and one goal call. The timing-only entry point `run_timing` runs the event source, the coarse pass and the crawl.

none. No candidates and no candidate list. This is the default and the setting of every main experiment arm.

motion. A free, deterministic pixel pseudo-signal. Each frame of one camera is converted to integer grayscale (BT.601 weights), box-averaged to a long side of 128 pixels in integer arithmetic, and differenced against the previous frame; the per-pixel mean difference is smoothed over 3 frames. Pauses are maximal runs of at least $\text{round}(0.3 \text{ fps})$ frames at or below the clip's 20th percentile of the smoothed signal, and each pause yields a `pause_start` at its first frame and a `pause_end` after its last, with confidence $1 - \bar{d}/\theta$ for mean difference $\bar{d}$ and threshold θ . Pauses that touch the first or last frame give no event there. Every step up to the final divisions uses integers, so identical frames give byte-identical events on any machine; the test suite pins the digests.

gripper. The V-lite L1 layer, unchanged except for two additive fields (the recovery candidates and their version). It reads the commanded and measured gripper positions (on the SO-101, `action[5]` and `observation.state[5]`), calibrated per dataset from development episodes only. A run is at least $\max(2, \text{round}(3 \text{ fps}/30))$ consecutive frames whose 3-frame smoothed command velocity exceeds 1% of the command range per frame; same-direction runs closer than 0.25 s merge. Each closing run gives a `close_start` and each opening run an `open_start`, with confidence 0.9. Low-confidence L1 candidates for the arm starting to move become `arm_move` events with confidence 0.3. For v1.1 L1 also proposes *recovery candidates* after every empty or aborted close that reopens: an `open_start` at the reopening onset (high confidence), and a `back_off` where the arm, after coming to rest, starts moving away from its pose at the close (low confidence; kept only if at least half of the next 0.5 s of displacement points away from that pose and the move starts at least 0.25 s after the reopening). On the 30 ArmNetBench development episodes of Section 5.1 this produced 90 re-open and 7 back-off candidates. With the gripper source, a close or open boundary that the model ties to a gripper event of the same type takes the signal's frame and is not crawled.

4.3 Scene inventory

The inventory call sees up to 8 evenly spaced frames, first and last included, and lists at most 10 objects that matter for labeling: every object the task description names or implies, everything the robot, humanoid or person touches, holds, moves, presses or uses as a tool, every place an object is put into or onto, and every look-alike of a handled object. Each object gets an identifier, a disambiguating name, aliases, a category and a point and box in the first frame. The inventory’s identifiers are used by the coarse pass, the scene facts and the goal. When the inventory is empty or invalid, targets are written in plain words and the scene-facts call is skipped.

4.4 The coarse pass

Input. In frames mode, the coarse pass sees $k = \min(\lfloor 2T + 0.5 \rfloor + 1, 48, N)$ evenly spaced frames of a clip of N frames, where $T = (N - 1)/\text{fps}$ is the time from the first to the last frame; the first and last frames are included, each frame with its caption. Clips longer than about 23.5 s hit the 48-frame cap and are sampled below 2 frames per second: a 30 s clip is read at 1.57 frames per second. In video mode it receives the clip itself as one native video part (an H.264 mp4 of at most 60 s) and answers in seconds, converted to frames by $\text{round}(t \cdot \text{fps})$. robolabel never encodes video (for E1, the experiment tools re-encoded each episode with PyAV as H.264 at a long side of 448 pixels), and the rate at which the provider samples the video is neither set nor recorded by the package.

Output. The model splits the whole clip into consecutive, non-overlapping segments. Each segment has a start and end frame, a `phase_text` in the model’s own words that names the object acted on, a `phase_class` from 16 classes (approach, grasp, transport, release, retract, press, pour, insert, fold, wipe, push, pull, rotate, open, close, other; “the phase list is a vocabulary, not a template”), an `end_event`, a target and destination, an attempt index, the phase’s own outcome, the attempt’s outcome, a failure type and an optional candidate identifier. The `end_event` is the kind of boundary at the end of the segment, the event that starts the next one: `close_start` (the fingers of a gripper or hand start to close), `open_start` (they start to open), `contact_start` and `contact_end` (a hand or held tool starts or stops touching an object or surface), or `other`. During development, GPT-6 Luna typed every grasp and release on an SO-101 episode as a contact event, which is physically defensible and would have scored as a miss. Before the prompts were frozen we added a two-sentence paragraph: when fingers close on an object to hold it the boundary is `close_start`, when they open to let it go it is `open_start`, and contact events are for touches in which the fingers neither close nor open. The prompt also states “There is no minimum number of segments. Never add a phase that you cannot see.”

Post-processing. A deterministic pass repairs and records every deviation: it converts and clamps frame values, swaps reversed spans, maps unknown enumerations to defaults, resolves object names to inventory identifiers, sorts the segments, closes gaps by extending the earlier segment and cuts overlaps at the later start, stretches the first segment to frame 0 and the last to the final frame, forces the last `end_event` to `other`, records each segment’s proposed end as `coarse_end_frame`, applies the gripper snap when a signal is present, and applies the failure convention (Section 4.6). An answer with no usable segment becomes a single “no output” segment covering the clip, which counts as wrong in every metric.

4.5 The crawl

The crawl refines, in time order, every boundary whose end event is a finger or contact event (`close_start`, `open_start`, `contact_start` or `contact_end`), at most 12 per episode and at most 3 calls per



Figure 3: One `open_start` boundary of PickPlace episode 0 as the crawl saw it (GPT-6 Luna, arm L-B of experiment E1). The coarse pass proposed onset frame 217. Stage 1 (top) showed 8 frames over ± 1 s and the model answered image 6 (frame 230). Stage 2 (bottom) showed 8 of the 10 frames from image 5 to image 6 (frames 221 to 230) and the model answered image 3 (frame 224). Green marks the truth, the onset of the opening run of the measured gripper position (frame 221), which the model never saw. Red marks the model’s pick in each stage.

boundary. It asks one narrow question per call over a window of frames and returns an onset frame: the first frame of the segment after the boundary.

Windows. Stage 1 takes 8 evenly spaced frames over ± 1 s around the coarse onset (at 30 fps, frames 120, 129, 137, 146, 154, 163, 171, 180 around onset 150). A window that would leave the clip is shifted back inside it at full width. Stage 2 takes every frame between the stage-1 pick and the image before it if they fit in 8 images, otherwise the finest even spacing that fits. All window arithmetic is in integers.

Question and answers. The question names the event type (“Which of these frames is the first where the fingers (grripper or hand) have started to close?”; for contact events “... where the hand or the tool touches the *paper*”, with the object taken from the neighboring phases), optionally the two phases the boundary separates, and the answer codes. The model replies with one integer: 2 to n means image k is the first to show the event (it began between images $k - 1$ and k); 0 means image 1 already shows it; 9 means the fingers, hand or tool are visible and it has not begun by image n ; -1 means it cannot be judged in these images.

Algorithm. A pick in stage 1 leads to stage 2 between the image before the pick and the pick, unless the two are adjacent frames. An answer of 0 or 9 shifts the stage-1 window by its own width earlier or later and asks once more; a pick in that window that agrees with the first answer continues to stage 2, and a contradiction keeps the stage-1 edge with a flag. In stage 1, an answer of -1 or a failed call leaves the coarse onset in place; later in the crawl, they keep the result reached so far. A stage-2 answer that contradicts stage 1 keeps the stage-1 result. A refined onset must lie strictly between the previous refined boundary and the next coarse boundary; a crossing keeps the coarse frame. Every call records the frames it saw, its answer and its cost, and every boundary records its coarse and refined frames, its flags (`crawl_edge`, `crawl_none`, `crawl_inconsistent`, `crawl_cross`, `crawl_failed`) and `boundary_source`: `crawl` when the crawl decided it. The repository’s `SCHEMA.md` lists every crawl flag and the rule behind it. Figure 3 shows one boundary as the model saw it.

4.6 Failures, attempts and recovery

An attempt is one try at a step: an approach and a close on an object, or an approach and a contact with a control or surface. Each phase’s `outcome` is its own result. In a missed grasp, the approach that reached the object is a success and the grasp is failed with `failure_type`: `missed_grasp`. A separate

`attempt_outcome` (success, failed or aborted) is copied onto every phase of the attempt, so a consumer can still drop whole failed attempts, and the per-segment `mistake` flag is true only on the phase that failed. The prompt states a recovery convention: the failed grasp ends when the fingers start to reopen (its `end_event` is `open_start`), a `retract` segment follows if the arm or hand backs off, still in the failed attempt, and then the approach of the next attempt begins. Post-processing enforces the failure convention: it keeps at most one failed phase per attempt (the one whose class fits its failure type), makes attempt indices non-decreasing, derives each attempt’s outcome from its phases, and writes one attempt record per attempt spanning its first to its last frame.

4.7 Goals, commands and activities without an end state

The goal call sees the first and last frame, the task description when there is one, the inventory, and the scene facts at the first and last keyframes; it never sees the segments, so the goal cannot copy their wording. It returns an `objective_text`, a `has_end_state` flag, a primary target and destination, and a list of requirements. Each requirement has a kind (object or robot end state), a predicate from 17 (`inside`, `on_top_of`, `touching`, `in_gripper`, `lifted`, `at_location`, `unchanged`, `activated`, `state`, `holding`, `gripper_open`, `gripper_closed`, `withdrawn`, `at_home_pose`, `near_object`, `tool_lifted`, `other`), an object and reference, a value, a status (required, incidental or unsure, with the kind of uncertainty: perception when the images cannot show the fact, intent when the task does not settle whether it is required), a basis, whether the clip achieved it, and where it is best judged. A robot or humanoid gets exactly three robot items (`holding`, `gripper open` or `closed`, and a position item); a person gets none. “Nothing that happened only during a failed attempt is a requirement”, and post-processing rejects object requirements that refer only to failed attempts.

State and command. The objective stays a statement of the end state (“The pink lego brick is inside the transparent box”), which can be checked against the last frame. A word-list check (101 imperative verbs, 88 ambiguous words that can start either a command or a state sentence, and state verbs such as “is”, “remains” and “ends”) rejects an objective that reads as a command and re-renders it from the requirements. The command for training prompts, `goal_command`, is rendered deterministically from the required object end states in their order: `inside` gives “Put *object* in *ref*”, `on_top_of` gives “Put *object* on *ref*”, `activated` gives “Press *object*” for a control and “Turn on *object*” otherwise, and `state` gives “Set *object* to *value*”; items are joined with “then” and robot items never appear. For the humanoid clip of Section 6.5 this gives “Put the slice of bread in the white toaster then press the toaster lever”. Six further forms (take out of, take off, turn off, move to, lift, pick up) exist behind a flag that is off by default, because the text becomes training input and each new verb needs a human sign-off.

No end state. `has_end_state` is false for activities meant to leave no object in a particular state or place, such as a dance, waving or a gesture. The objective then describes the activity as a statement, and no object end-state requirement may appear; a rule checks this.

4.8 Scene facts, checks and routing

Scene facts are read at keyframes placed at frame 0, at each refined boundary onset (typed boundaries first) and at the last frame, at most 8. For each image the model lists the visible and partly visible objects, the object held in a gripper or hand, and up to 4 relations (`inside`, `on top of`, `touching`) among the task objects. Thirteen rules then check the labels (listed in the repository’s `SCHEMA.md`). Five need a robot signal (for example, every missed grasp that the signal detects must be matched by a failed segment) and are marked not applicable without one. Three are specific to v1.1: every crawl pick lies inside a window the model saw, no boundary moved by the crawl crosses a neighbor, and `has_end_state: false` implies no object end

state. The episode’s risk is the share of applicable rules that failed. An episode is routed to review when a call failed, when there is no output, when an object end state is uncertain for perceptual reasons and no signal decides it, or, with a signal, when a claim contradicts it.

4.9 Infrastructure for trustworthy measurement on a budget

The pipeline is one part of the system; the rest exists so that small, cheap experiments can be trusted.

Provider. All model calls of the V-lite and v1.1 experiments go through one OpenRouter provider file. A call is a system message, text and image (or video) parts, and a JSON schema sent in strict structured-output mode, falling back to a JSON object and then to plain text with the schema in the prompt when a model refuses strict mode. The schemas are written to work on every provider (no nulls, every property required, no `$ref`, `oneOf` or numeric limits; limits are stated in the prompt and enforced after parsing). No sampling temperature is sent. An invalid answer gets one repair retry with the validation errors; a response cut at the token limit gets one retry with the limit doubled, up to 16,000 tokens.

Spend guard. Before every HTTP attempt, retries included, the provider reserves the attempt’s worst-case cost: the estimated input tokens (text characters divided by 3.5, plus a measured per-image token count times 1.2) at the input price plus the maximum output tokens at the output price. The guard refuses a reservation that would take the committed total (reconciled spend, plus unreconciled spend, plus open reservations) past the run cap, past a per-bucket cap, or below a floor on the account balance. The reservation is replaced by the provider-reported cost when the response arrives. A timeout, dropped connection or server error keeps its reservation as spent until it can be reconciled from the provider’s generation record, and reservations still open after a crash count as spent. Every event is appended to a ledger and fsynced, one process at a time may reserve against a ledger, and every 10 calls or 5 minutes the ledger is compared with the key’s own usage reported by the provider, pausing and then stopping paid calls on a persistent discrepancy. A refused call has status `refused` and the episode continues without it.

Cache and receipts. A response is cached under the SHA-256 of the provider, model, full prompt text (captions included), generation settings and the ordered hashes of the images and video, so reruns are free and any prompt change forces a new call. Every call writes a receipt with its inputs’ hashes, token counts, cost, latency, status and response text, and never the key, headers or image bytes.

Held-out guard. Every command that would run a system on, display or score an episode first passes a guard that refuses held-out episodes unless given the hash of a committed preregistration, and logs any such access. v1.1 adds a family of short clips outside every benchmark split, keyed `C/<clip id>`, which the guard accepts only when listed in the run’s clip list. No held-out episode was touched in any v1.1 experiment.

Determinism and tests. The event sources, the frame plan, the crawl windows, post-processing, the command renderer, the checks and the parquet writer give byte-identical output on identical input, and tests pin them. The repository has 753 passing tests (7 skipped), none of which call a model or use the network. The prompts that E1 ran with are frozen by hash: the runner refuses to start if the rendered requests for a fixed synthetic episode differ from the frozen state.

4.10 Output

The output is a long-format parquet file with one row per record: the episode’s metadata, its subtasks and coarse subtasks, attempts, scene facts, requirements and checks. The schema is `robolabel/`

Table 3: E1 data. Truth events are onsets of closing and opening runs of the gripper, found by the deterministic run detector of Section 4.2. The offset is the median of the measured onset minus the commanded onset over same-type pairs within 1 s.

	PickPlace	ArmNetBench
Dataset	svla_sol101_pickplace	ArmNetBench v0.1 (SO-101)
Episodes (development)	30	30
Camera, rate	up, 30 fps, 640 × 480	front, 20 fps, 1024 × 576
Truth events, measured position	127 (60 close, 67 open)	293 (146 close, 147 open)
Truth events, command	129 (61, 68)	314 (154, 160)
Events per episode, median (range)	4 (4 to 5)	6.5 (4 to 27)
Measured minus commanded onset	+4 frames (0.13 s)	+3 frames (0.15 s)

annotations/v7 with 11 optional v1.1 columns, among them the end event, the coarse end frame, the crawl calls per boundary, the attempt outcome, `has_end_state` and `goal_command` (documented in the repository’s `SCHEMA.md`). Files from every earlier schema version still read. Each run also returns a plain-JSON view record with the crawl log, step statuses and cost.

5 Evaluation

We ran three experiments, designed before any model call and executed in one unattended run on September 27 and 28, 2026. E1 measures boundary timing against truth that the models never see. E2 runs the full pipeline with six models on six short clips from six domains. E3 reads the E2 outputs for signs that the labels were forced into a pick-and-place shape. One rater scored every E2 output on a blind page; six of the 36 main-arm cards lost their blinding before the ratings were saved (Section 6.4). Every v1.1 result is on development data, and no held-out episode was touched; the v6 held-out numbers of Section 3.1 date from June 2026.

5.1 E1: boundary timing against hidden gripper signals

Data. E1 uses the development episodes of two datasets recorded with the SO-101 arm [30] (Table 3). PickPlace is `lerobot/svla_sol101_pickplace`: the 30 tune episodes of a frozen 50-episode split, each a short pick and place of a brick into a box, filmed by the camera named `up` at 30 frames per second. ArmNetBench is ArmNetBench v0.1, `armnet/armnetbench_v01_lerobot_sol101` on the Hugging Face Hub [29]: 30 development episodes drawn with a fixed seed, 15 of the task “Put the eye drops into the basket” and 15 block stacking, each task with 4 teleoperated successes, 7 policy failures and 4 policy successes, filmed by the `front` camera at 20 frames per second. The policy failures contain many retries: the 14 policy-failure episodes hold 214 of ArmNetBench’s 293 truth events. Every model call sees one camera and the dataset’s task text; the gripper channels are removed from the episode object before any request is built.

Truth. The primary truth is the onsets of the closing and opening runs of the *measured* gripper position (`observation.state[5]`), found by the same detector as the gripper event source, with a per-dataset calibration from development episodes. The measured position is what a camera can see: the fingers move when the position changes. The secondary truth is the onsets of the *commanded* position (`action[5]`), which leads the measured one by a median of 3 to 4 frames on both datasets. Every run counts, including openings during the approach and re-closes after a release; a run already in progress at frame 0 is excluded (25 ArmNetBench closes). The truth never enters a prompt, and the runner never reads the truth files. We

Table 4: E1 arms. “Frames” is the coarse pass on frames at 2 per second (at most 48); “video” is native video as the provider samples it. “Motion” adds the motion candidates of Section 4.2. G-BG ran until its budget bucket was spent, 15 of 20 episodes.

Arm	Coarse model	Input	Candidates	Crawl model	Episodes
L-A	GPT-6 Luna	frames	none	none	60
L-B	GPT-6 Luna	frames (L-A’s)	none	GPT-6 Luna	60
L-D	GPT-6 Luna	frames	motion	GPT-6 Luna	60
G-A	Gemini 3.8 Flash	frames	none	none	20
G-V	Gemini 3.8 Flash	video	none	none	20
G-B	Gemini 3.8 Flash	frames (G-A’s)	none	GPT-6 Luna	20
G-VB	Gemini 3.8 Flash	video (G-V’s)	none	GPT-6 Luna	20
G-BG	Gemini 3.8 Flash	frames (G-A’s)	none	Gemini 3.8 Flash	15 of 20

score both truths because they encode two defensible definitions of when a grasp starts, and a labeler has to pick one.

Arms. Table 4 lists the eight arms. All use the same frozen v8 prompts. GPT-6 Luna, the cheapest model in our sweep, proposes boundaries for the L arms on all 60 episodes; Gemini 3.8 Flash proposes them for the G arms on 20 interleaved episodes (10 per dataset). A Gemini crawl call was expected to cost about 20 times a Luna call (it cost about 26 times), because Gemini reads each image as about 1,088 input tokens against about 133 for Luna, so the main crawl comparisons use the Luna crawl on both proposers, and G-BG measures what a Gemini crawl buys. An arm that shares a coarse pass with another reuses it through the response cache and never calls the coarse model again, so the three crawl contrasts (L-B against L-A, G-B against G-A and G-BG against G-B) isolate the crawl; the input and hint contrasts compare separate coarse calls and also include the coarse model’s run-to-run variation. The coarse pass and the crawl are the only calls in E1; the coarse pass names targets in plain words.

Metrics. Only boundaries typed `close_start` or `open_start` are scored, as onset frames, against truth onsets of the same type. Contact boundaries are crawled and paid for but not scored, since the SO-101 has no contact sensor. For each arm we score the coarse boundaries and the refined boundaries. The matching and the bootstrap are implemented in the package’s `eval` module [35], and the E1 scorer that applies them is part of the experiment tools:

- **T1 precision, recall and F1** at tolerances τ of 3, 5 and 10 frames, from an optimal one-to-one matching per type that first maximizes the number of pairs within τ and then minimizes their summed absolute error, with counts summed over the two types (so $F_1 = 2M/(m + n)$ for M matches, m predictions and n truth events), micro-averaged within a dataset and macro-averaged across the two datasets (“pooled”). At 30 fps, 5 frames is 0.17 s; at 20 fps, 0.25 s.
- **T2-MAE**, the mean absolute frame error over the typed pairs matched at $\tau = 10$. It counts matched pairs only, so it is read together with the pair count and the miss rate.
- **Median signed offset** (prediction minus truth) over optimal typed pairs within 1 s, and an **offset-corrected MAE** after shifting every prediction by minus that offset. The shift is fitted and scored on the same episodes, so this number flatters itself and is reported only as a diagnostic.
- **Missed within 1 s**, the share of truth events with no prediction of the same type within 1 s.

An episode whose coarse call failed counts as having no predictions; no episode is dropped.

Statistics. Each contrast is a paired difference between two arms on the same episodes, with a 95% percentile interval from a cluster bootstrap over episodes ($B = 10,000$), stratified by dataset, recomputing each metric from counts on every replicate. The seed of contrast i is $20261001 + i$, with i its position in the list of contrasts fixed before the run. Every contrast gets one of three outcome words, fixed before the run [26]:

- **clear difference** when the interval excludes 0;
- **no clear difference** when it includes 0 and its half-width is at most the minimum effect of interest (MEI);
- **inconclusive** otherwise, or when the subset has fewer than 15 episodes.

The MEI is 0.05 for P, R and F1, 1 frame for T2-MAE and 0.05 for the miss rate. Five contrasts carry outcome words (L-B–L-A, L-D–L-B, G-B–G-A, G-V–G-A, G-VB–G-B); G-BG–G-B is descriptive. We apply no correction for the number of contrasts: E1 runs on development data and is exploratory by construction.

5.2 E2: six models on six domains

Clips. E2 runs the full pipeline on six clips (Table 5), chosen to cover a robot arm, a humanoid, egocentric and exocentric human video, a robot hand using a tool, and an activity without an object end state. Three clips have truth for some boundaries: the SO-101 clip (the gripper onsets of the E1 truth; the clip is ArmNetBench episode 1821, one of the E1 episodes), the humanoid clip (right-hand closure onsets from the Dex3 hand joints) and the drawing clip (pencil contact from the simulator’s contact force, at least 0.015 N for at least 3 frames). The drawing clip was rendered in MuJoCo [31] from the author’s statebench project [34]. The pencil tip hovers within 2 mm of the paper for about 10 frames before each contact, so a visual `contact_start` is ambiguous by about that much. The clips are H.264 files cut with PyAV. Only the robot and humanoid clips pass a task string to the prompts.

Models. Table 6 lists the six models, all called through OpenRouter with identical settings: reasoning at effort low with the reasoning text excluded from the response (it is still billed), strict JSON schemas, no temperature, and per-call token limits of 3,000 (inventory), 8,000 (coarse), 2,500 (crawl), 6,000 (facts) and 5,000 (goal). Three further models from the V-lite sweep were dropped before the run because of latency or invalid output (Section 3.2).

Arms. The main arm is the default v1.1 pipeline: no event source, frames, and the crawl by the same model. It runs for every model on every clip (36 outputs). Five extras run right after the matching main arm: the gripper event source on the SO-101 clip for Luna, Gemini and Sol, and native video on the SO-101 and drawing clips for Gemini. The gripper extras see the signal by design, so they are compared with their own main arm and never scored against gripper truth. All 41 outputs were rated.

Measures. On the clips with truth, we score the typed boundaries of the main arms and the video extra as in E1, for the event types the clip’s truth has. On every clip we record validity, routing, cost, wall time and crawl statistics. Every E2 cell is a single clip, so every E2 number is descriptive: it carries no interval and no outcome word.

Table 5: E2 clips, in the order every model ran them. Frames sent is the coarse pass’s effective rate; clips longer than about 23.5 s reach the 48-frame cap.

Clip	Content	Length, rate	Frames sent	Truth	License
SO-101 arm	ArmNetBench episode 1821: eye drops into a basket after four missed grasps and an aborted close; task text given	23.7 s, 20 fps	1.99/s	6 closes, 6 opens (gripper)	Apache-2.0
Dance	son jarocho couple dancing at a fandango; nothing handled by the dancers	18.9 s, 30 fps	2.01/s	none	CC BY-SA 4.0
Sim drawing	Kinova Gen3 arm with a Shadow Hand drawing three pencil strokes (MuJoCo)	37.4 s, 20 fps	1.26/s	3 contact starts, 3 ends	author’s
Egocentric	head-mounted view of folding laundry	30.0 s, 30 fps	1.57/s	none	CC BY 4.0
Humanoid	Unitree G1 with Dex3 hands putting bread in a toaster; task text “toasted bread”	20.7 s, 30 fps	1.99/s	3 closes, 3 opens (right hand)	Apache-2.0
Third person	close view of hands washing a bowl and plates; on-screen titles	30.0 s, 30 fps	1.57/s	none	CC BY-SA 4.0

Table 6: Models in E2, with list prices in USD per million input and output tokens on September 27, 2026, and the input tokens one image costs. All but Muse list a batch price at half the list price; batch was not used. Gemini’s price is promotional until the end of 2026.

Model	OpenRouter id	Input	Output	Tokens per image
GPT-6 Luna	openai/gpt-6-luna	0.10	0.50	133
Gemini 3.8 Flash	google/gemini-3.8-flash	0.75	3.75	1,088
GPT-6 Sol	openai/gpt-6-sol	2.00	10.00	133
Muse Spark 1.3	meta/muse-spark-1.3	1.25	4.25	147
Claude Opus 5.5	anthropic/claude-opus-5.5	4.00	20.00	148
GPT-6 Astra	openai/gpt-6-astra	10.00	50.00	133

5.3 E3: do the labels stay general?

Each clip is tagged beforehand with whether pick and place fits it (yes: the SO-101 and humanoid clips; partly: drawing, egocentric and third person; no: the dance) and whether it has an object end state (all but the dance). On the “partly” and “no” clips we measure the *forcing rate*, the share of refined segments whose `phase_class` is `approach`, `grasp`, `transport`, `release` or `retract`, and the share whose `phase_text` contains one of those words. On the dance, `has_end_state` must be false and no object end state may appear. The rater also judged whether the labels fit the activity.

5.4 Rating protocol

One rater, the author, scored every E2 output on a static review page. Each clip has one tab with the video, a timeline player and one card per output, lettered in an order shuffled with a secret salt. A card shows the labels only: the phases with their spans, texts, classes and outcomes, the attempts, and the goal with its command. It never shows a model name, an arm, a cost, a time or any field that reveals how a label was made

Table 7: E1 per arm, pooled over the two datasets (macro average of each dataset’s micro value), against the measured gripper onset. “Refined” rows are after the crawl; arms without a crawl have one row. MAE is over pairs matched at 10 frames (pair count in brackets); the offset is the median of prediction minus truth, in frames (pooled rows average the two datasets’ medians); OC-MAE is the MAE after removing that offset, fitted on the same episodes. Cost is the whole method per episode, a reused coarse pass included at its original price.

Arm	Boundaries	F1@3	F1@5	F1@10	MAE (pairs)	Offset	OC-MAE	Missed@1s	\$ / ep.
L-A	coarse	0.126	0.175	0.254	4.14 (62)	−1.0	4.09	0.724	0.0011
L-B	refined	0.129	0.252	0.304	4.21 (72)	+4.0	2.61	0.716	0.0033
L-D	coarse	0.153	0.200	0.271	4.01 (64)	0.0	3.12	0.730	
L-D	refined	0.089	0.271	0.333	4.54 (76)	+4.5	2.25	0.716	0.0030
G-A	coarse	0.123	0.263	0.420	5.15 (34)	+1.0	3.98	0.645	0.0245
G-B	refined	0.156	0.378	0.428	4.49 (34)	+4.25	1.39	0.650	0.0260
G-V	coarse	0.080	0.144	0.215	4.22 (23)	−12.0	4.13	0.656	0.0046
G-VB	refined	0.048	0.136	0.223	4.75 (24)	−4.25	3.11	0.651	0.0062
G-BG [†]	coarse	0.112	0.203	0.395	5.62 (26)	+2.0	3.98	0.655	
G-BG [†]	refined	0.224	0.345	0.446	4.67 (29)	+3.25	2.87	0.661	0.0641

[†] 15 episodes (8 PickPlace, 7 ArmNetBench); all other G arms 20 (10 per dataset), L arms 60 (30 per dataset).

(the event source, coarse mode and rate, crawl model and calls, boundary sources, coarse frames or the crawl log); an automatic check of the built site tests for these fields and for model names. In quick mode, the rater gives each card an overall score from 1 to 5 (1 unusable or wrong, 2 major errors, 3 usable after several edits, 4 minor edits, 5 correct as is), whether it is usable as a training label (yes, with edits, no), and whether the labels fit the activity (yes, partly, no), and picks one best card per clip. The key that maps letters to arms is kept apart from the site and checked by fingerprint when ratings are imported. The rating took 19 minutes of active time for 41 cards. Six of the 36 main-arm cards were no longer blind when the ratings were saved; Section 6.4 reports the ratings with and without them.

An earlier round, run the day before on the V-lite pipeline, used the same rater, page and scale on four development episodes (three SO-101 episodes and one LIBERO simulation episode), with nine models, the old prompts and a signal-only arm; its outputs had the gripper signal as boundary candidates (Section 3.2). We report both rounds, because together they are the only human judgment of these models on this task that we have, and because they disagree.

6 Results

6.1 E1: the crawl, the input and the hints

Table 7 gives every arm’s pooled results against the measured onset, and Table 8 the paired contrasts against both truths. Figure 4 shows the contrasts and Figure 5 the change the crawl makes within each arm. All 275 runs finished (550 scored rows, coarse and refined); G-BG stopped by design at 15 of its 20 episodes.

The crawl helps against the measured onset. On Luna’s proposals, the crawl raised typed boundary F1 at 5 frames from 0.175 to 0.252, a clear difference of +0.077 [0.036, 0.119]. F1 at 10 frames also rose clearly (+0.049 [0.019, 0.082]), as did recall at 5 frames (+0.058 [0.030, 0.088]); the change at 3 frames was inconclusive (+0.003 [−0.052, 0.059]). The matched-pair MAE showed no clear difference (+0.07 frames), and neither did the miss rate. On Gemini’s proposals the same Luna crawl raised F1 at 5 frames from 0.263 to 0.378, a clear difference of +0.115 [0.028, 0.212]; the MAE change was inconclusive. A crawl by Gemini itself gave nearly the same F1 at 5 frames as the Luna crawl on the 15 episodes both ran (0.345 against 0.346;

Table 8: E1 contrasts: paired differences (first arm minus second) with 95% cluster bootstrap intervals and the pre-declared outcome word (C: clear difference; N: no clear difference; I: inconclusive; D: descriptive). Indented rows restrict the first contrast to one dataset. Higher F1 is better; lower MAE and lower miss rate are better. MAE is in frames.

Truth	Contrast	n	F1 at 5 frames		MAE		Missed within 1 s	
			Diff. [95% CI]		Diff. [95% CI]		Diff. [95% CI]	
Measured	L-B – L-A	60	+0.077 [0.036, 0.119]	C	+0.07 [-0.73, 0.88]	N	-0.008 [-0.020, 0.000]	N
	PickPlace	30	+0.184 [0.118, 0.257]	C	-1.12 [-2.52, 0.30]	I	-0.016 [-0.040, 0.000]	N
	ArmNetBench	30	-0.030 [-0.075, 0.012]	N	+1.26 [0.42, 2.03]	C	0.000 [0.000, 0.000]	N
	L-D – L-B	60	+0.019 [-0.022, 0.059]	N	+0.34 [-0.29, 0.99]	N	+0.000 [-0.022, 0.022]	N
	G-B – G-A	20	+0.115 [0.028, 0.212]	C	-0.66 [-1.64, 0.39]	I	+0.005 [0.000, 0.014]	N
	G-V – G-A	20	-0.120 [-0.204, -0.042]	C	-0.93 [-3.08, 1.60]	I	+0.011 [-0.069, 0.075]	I
	G-VB – G-B	20	-0.242 [-0.304, -0.188]	C	+0.26 [-0.17, 0.76]	N	+0.001 [-0.074, 0.061]	I
	G-BG – G-B	15	-0.000 [-0.076, 0.077]	D	-0.07 [-0.80, 0.74]	D	0.000 [0.000, 0.000]	D
Commanded	L-B – L-A	60	-0.060 [-0.102, -0.017]	C	+1.60 [0.85, 2.39]	C	-0.008 [-0.020, 0.002]	N
	L-D – L-B	60	-0.026 [-0.060, 0.007]	N	-0.16 [-0.98, 0.57]	N	+0.006 [-0.013, 0.025]	N
	G-B – G-A	20	-0.177 [-0.240, -0.109]	C	+1.60 [0.37, 2.85]	C	+0.014 [0.000, 0.022]	N
	G-V – G-A	20	-0.032 [-0.117, 0.068]	I	-1.23 [-2.45, 0.21]	I	-0.011 [-0.085, 0.043]	I
	G-VB – G-B	20	+0.038 [0.007, 0.081]	C	+0.15 [-1.06, 1.12]	I	-0.028 [-0.090, 0.025]	I
	G-BG – G-B	15	+0.148 [0.083, 0.210]	D	-1.58 [-2.40, -0.63]	D	-0.011 [-0.024, 0.000]	D

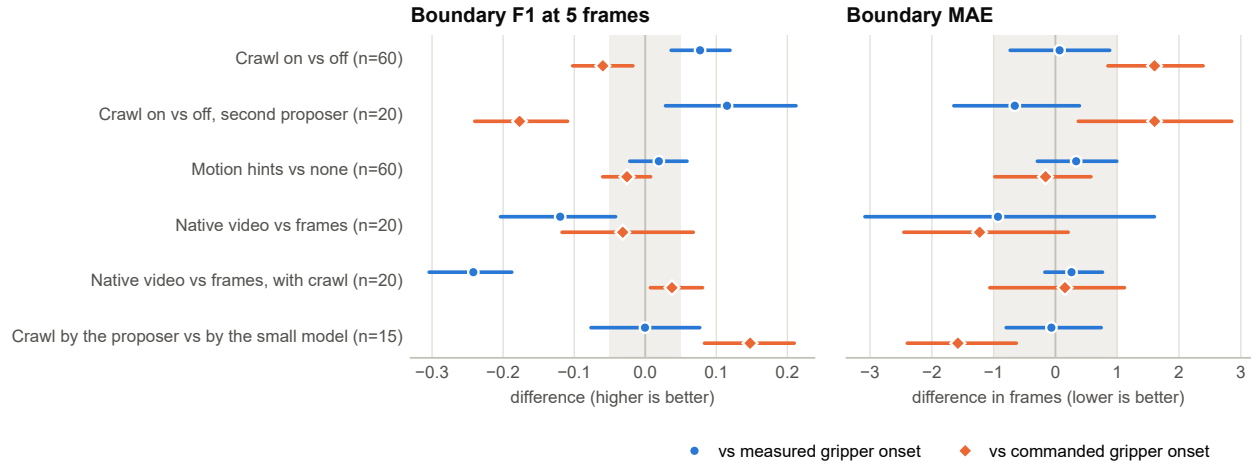


Figure 4: E1 contrasts, pooled over both datasets, with 95% cluster bootstrap intervals, scored against the measured (blue) and the commanded (orange) gripper onset. The gray band is the minimum effect of interest (0.05 for F1, 1 frame for MAE). The crawl helps against the measured onset and hurts against the commanded one; native video is worse than frames against the measured onset.

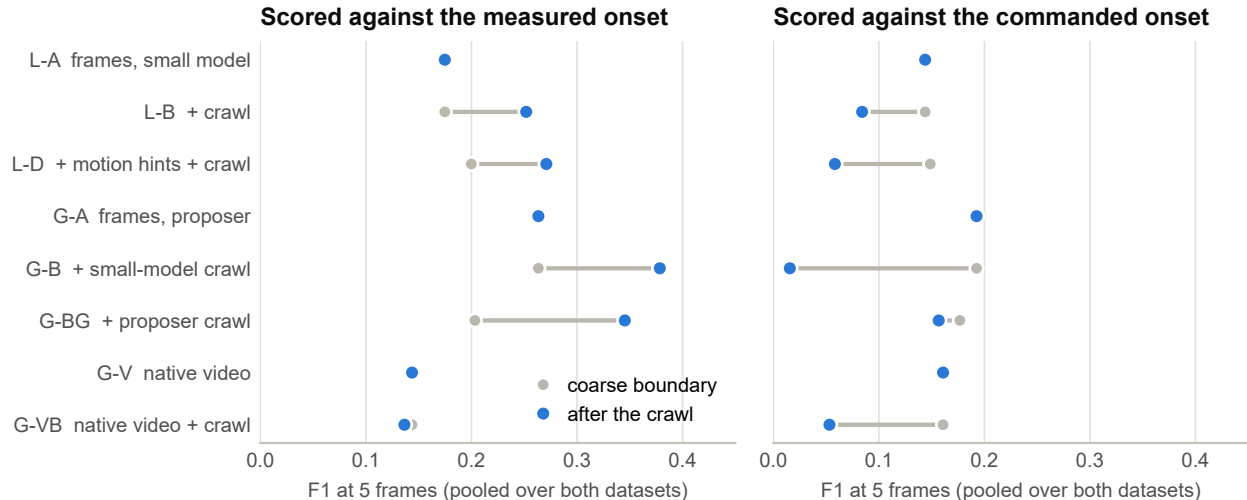


Figure 5: F1 at 5 frames for each E1 arm before (gray) and after (blue) the crawl, against the two truths. Arms without a crawl show one point. G-BG ran 15 episodes, the other G arms 20, the L arms 60.

descriptive, interval -0.076 to $+0.077$), at 27 times the crawl cost per episode ($\$0.039$ against $\$0.0015$).

The gain is one dataset’s. On PickPlace alone the crawl raised F1 at 5 frames by $+0.184$ $[0.118, 0.257]$, and precision at 5 frames by $+0.293$ $[0.186, 0.407]$. On ArmNetBench alone it made no clear difference at 5 frames (-0.030), a clear loss at 3 frames (-0.059 $[-0.102, -0.026]$) and a clearly worse MAE ($+1.26$ frames $[0.42, 2.03]$). Boundary by boundary, the crawl moved 40 of Luna’s 58 typed PickPlace boundaries closer to the nearest truth onset and 14 further; on ArmNetBench it moved 13 closer and 28 further. Figure 6 shows why. On PickPlace the coarse boundaries sat early (median -7 frames) and the crawl brought them to $+2.5$; on ArmNetBench they already sat close (median $+1$) and the crawl pushed them to $+4$. The pooled clear difference is therefore a result on `svla_sol01_pickplace`, and we do not claim it for both datasets.

The crawl hurts against the commanded onset. Scored against the command, the same crawl lowered F1 at 5 frames by -0.060 $[-0.102, -0.017]$ and raised the MAE by 1.60 frames $[0.85, 2.39]$, both clear differences; on Gemini’s proposals the losses were -0.177 $[-0.240, -0.109]$ and $+1.60$ frames $[0.37, 2.85]$. The crawl’s refined onsets sit a pooled median of 4 frames after the measured onset and 7.25 frames after the commanded one (arm L-B), which falls outside a 5-frame tolerance and inside a 10-frame one: against the command, F1 at 10 frames still rose clearly ($+0.053$ $[0.011, 0.096]$). Much of the error is a constant offset. Against the measured onset, after subtracting each dataset’s median offset, the MAE of the refined boundaries is 2.6 frames for L-B and 1.4 for G-B, against 4.1 and 4.0 for the coarse boundaries; against the command, it is 3.3 and 1.5. That shift was fitted on the scored episodes and has not been tested on others.

Native video scores below frames. With Gemini’s native video input, as the provider samples it, F1 at 5 frames was 0.144 against 0.263 for frames at 2 per second, a clear difference of -0.120 $[-0.204, -0.042]$; with the crawl on both, -0.242 $[-0.304, -0.188]$. The video proposals sat early, a pooled median offset of -12 frames (-21 on PickPlace), and on PickPlace the video arm placed no `close_start` within 10 frames of a true close. robolabel neither sets nor records the rate at which the provider samples video. A debug check measured about 70 input tokens per second of video (2,327 tokens for 10.1 s, 3,615 for 28.5 s), which is consistent with about one frame per second at low resolution; we did not verify the rate. Against the commanded onset the video contrast became inconclusive, and video plus crawl came out ahead of frames

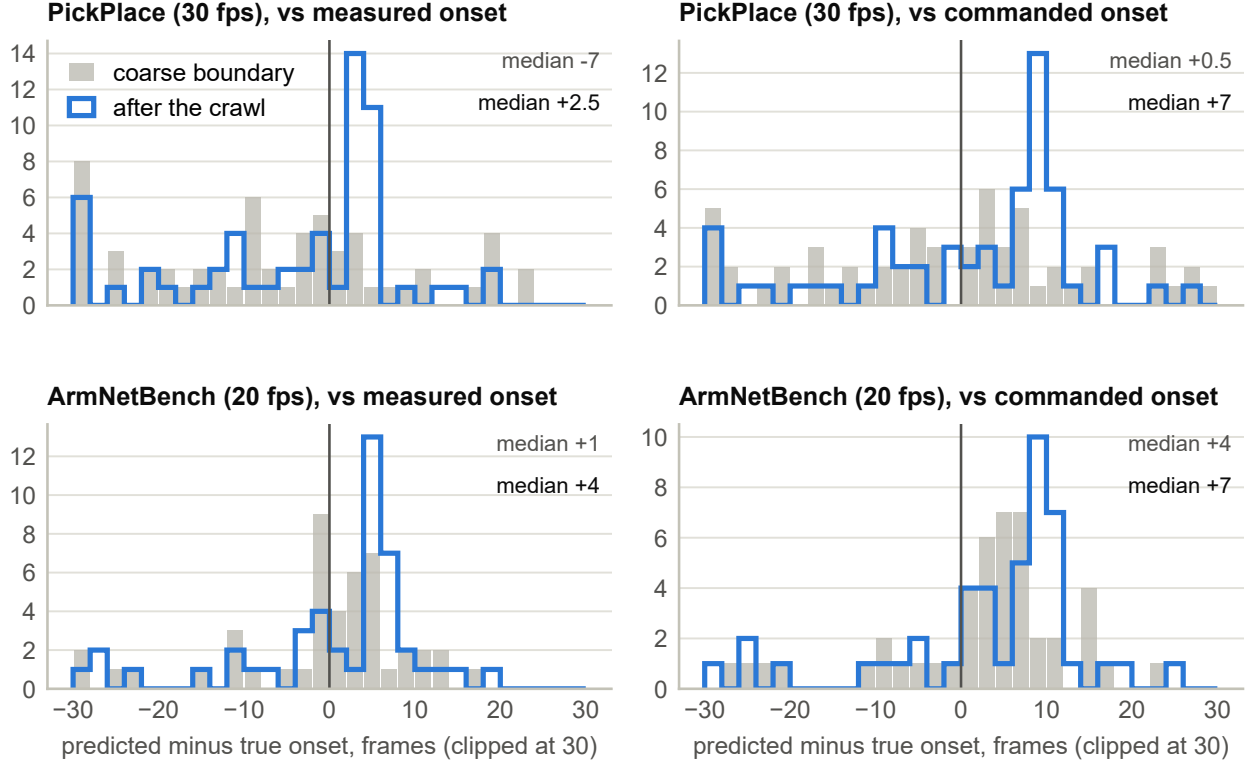


Figure 6: Signed error of arm L-B’s typed boundaries (Luna proposals, Luna crawl), predicted onset minus the nearest true onset of the same type, before (gray) and after (blue) the crawl. Left: against the measured gripper onset; right: against the commanded onset, which comes 3 to 4 frames earlier. On PickPlace the crawl moves early boundaries to a few frames after the measured onset; on ArmNetBench it moves boundaries that were already close to a few frames later.

plus crawl (+0.038 [0.007, 0.081]), which is consistent with the video proposals’ early bias partly cancelling the crawl’s late one.

Motion hints make no clear difference. Adding the free motion candidates to Luna’s coarse pass (L-D against L-B) made no clear difference in F1 at 5 frames (+0.019 [-0.022, 0.059]), MAE or misses. On PickPlace alone it gained at 5 frames (+0.071, clear) and lost at 3 frames (−0.088, clear).

Omission dominates. Every arm leaves 64.5 to 73.0% of true gripper events without a same-type prediction within 1 s, and the crawl does not change that (it moves boundaries and never adds one). The median distance from a true event to the nearest same-type prediction sits at the 1 s cap in every arm and dataset. Three causes contribute. First, the truth on PickPlace includes an opening during the approach and a close after the release, which a pick-and-place phase list does not mark. Second, on ArmNetBench the coarse passes fold retries into one attempt: on the 14 policy-failure episodes Luna proposed 13 typed events against 214 true events, and missed 94.9% of them. Third, the coarse pass on ArmNetBench proposed so few typed boundaries (45 against 293 true events for Luna) that even perfect placement would reach a recall of at most 0.15.

Crawl behavior and cost. Over its 60 episodes, arm L-B made 410 crawl calls (6.8 per episode) on 213 typed boundaries, 110 of them contact events that E1 does not score: 205 were crawled and 8 skipped at the 12-boundary cap, all on ArmNetBench. Its flags were 31 `crawl_inconsistent`, 20 `crawl_cross`,

Table 9: E2 main arm per model, six clips each. Cost is the whole pipeline per clip at list prices; the cost per 1,000 clips is 1,000 times the mean cost per clip, and “batch” applies the catalog’s batch price ratio (Muse lists none). Wall time is per clip and is only a rough speed measure, because the first pass ran all six models at once. Output rate is the median over calls of output tokens (reasoning included) per second.

Model	\$ / clip	\$ / 1,000 clips		Wall	Output	Crawl calls	
	median	list	batch	median	tok/s	per clip	Routed
GPT-6 Luna	0.006	5.71	2.86	66 s	62.1	8.7	2 of 6
GPT-6 Sol	0.094	93.29	46.64	64 s	37.1	6.2	1 of 6
Muse Spark 1.3	0.091	91.60	91.60	206 s	61.8	7.3	2 of 6
Claude Opus 5.5	0.149	157.23	78.61	46 s	20.5	3.7	2 of 6
Gemini 3.8 Flash	0.154	153.23	76.62	130 s	54.7	10.0	1 of 6
GPT-6 Astra	0.522	536.89	268.44	92 s	18.1	10.0	2 of 6

4 `crawl_edge` and 4 `crawl_none`; no pick in any arm fell outside the window the model saw. Luna’s coarse pass alone cost \$1.10 per 1,000 episodes; with the crawl, \$3.26 per 1,000 episodes at list prices (\$1.63 at the listed batch price, which we did not use), with a median wall time of 33 s per episode. The Gemini frame arms cost \$25 to \$26 per 1,000 episodes with or without the Luna crawl and \$64 with the Gemini crawl; the native-video arms cost \$4.60 and \$6.20. E1 cost \$1.58 in all.

6.2 E2: six models on six clips

All 41 E2 jobs finished with schema-valid output; none crashed, and 459 structured calls were made, a few of them answered from the response cache. Table 9 summarizes cost, speed and crawl use per model.

Cost and speed. The median cost per clip spans a factor of 84, from \$0.006 for Luna to \$0.522 for Astra; on average Astra cost 5.8 times as much as Sol per clip, where the budget plan had assumed 5, and Opus 1.7 times, where the plan had assumed 2. Opus typed the fewest boundaries as finger or contact events (11 over six clips), so it made the fewest crawl calls (3.7 per clip); Gemini typed the most (33) and crawled 30, reaching the 12-boundary cap on the egocentric clip, and Astra typed and crawled 29. Muse was the slowest per clip despite a high output rate; the model configuration notes that it reasons at length even at effort low.

Routing. The checks routed 13 of the 41 outputs to review. Every main-arm route was an end-state item marked unsure for perceptual reasons that no signal could decide: whether the gripper ended open on the SO-101 clip (all six models), whether the hand ended closed on the drawing clip (Luna, Muse and Astra), and the end state of one dish on Opus’s third-person output. The two gripper extras that routed did so because a claim contradicted the signal. Routing follows these conditions rather than a risk threshold: four routed outputs had risk 0, while the highest-risk main output (Gemini on the third-person clip, risk 0.33) was not routed. No output had a failed call.

Accuracy on the clips with truth. Table 10 gives the typed boundary scores on the three clips with truth. Each cell is one clip, so the numbers are descriptive. On the SO-101 clip every model proposed between 2 and 6 typed onsets against 12 true ones and at most 3 of the 6 closes; the four models that proposed an `open_start` near frame 208 placed it 25 to 30 frames before the measured opening at frame 236. After the crawl, the `close_start` onsets sat 0 to 9 frames after the truth, the late visible onset E1 measured, and the crawl moved 16 of the 26 typed boundaries further from the truth and 6 closer. Luna typed 6 of its 8 boundaries on this clip as contact events and labeled its two failed grasps as `push`. On the drawing clip, no

Table 10: E2 typed boundary F1 on the three clips with truth, coarse and after the crawl (refined), at 5 and 10 frames. Truth: SO-101 12 gripper onsets (20 fps); humanoid 6 right-hand onsets (30 fps; left-hand predictions count as unmatched); drawing 6 pencil contact onsets (20 fps). One clip per cell: descriptive only.

Model	SO-101 arm			Humanoid			Sim drawing		
	Pred.	F1@5 coarse / ref.	F1@10 ref.	Pred.	F1@5 coarse / ref.	F1@10 ref.	Pred.	F1@5 coarse / ref.	F1@10 ref.
GPT-6 Luna	2	0.14 / 0.00	0.14	1	0.00 / 0.00	0.00	0	0.00 / 0.00	0.00
GPT-6 Sol	5	0.35 / 0.35	0.35	2	0.00 / 0.00	0.00	2	0.00 / 0.00	0.00
Muse Spark 1.3	5	0.35 / 0.00	0.35	4	0.00 / 0.40	0.40	2	0.00 / 0.00	0.00
Claude Opus 5.5	3	0.40 / 0.40	0.40	4	0.20 / 0.00	0.20	2	0.00 / 0.00	0.00
Gemini 3.8 Flash	6	0.22 / 0.33	0.56	2	0.00 / 0.00	0.25	2	0.00 / 0.00	0.00
GPT-6 Astra	5	0.35 / 0.35	0.47	4	0.40 / 0.00	0.20	0	0.00 / 0.00	0.00
Gemini, video	5	0.12 / 0.24	0.24				4	0.00 / 0.20	0.20

model placed a single pencil contact within 10 frames of the truth: the crawled contact boundaries were 37 to 96 frames away, far beyond the pencil’s 10-frame hover ambiguity. Gemini’s native-video extra matched one contact end within 1 frame. On the humanoid clip, refined F1 at 5 frames was 0 for five of the six models.

Goals and commands. Every output had a state objective. The rendered command was empty on 14 of the 41 outputs: all six dance outputs (as intended, since the dance has no end state), all six egocentric outputs, and Astra’s drawing and third-person outputs. The renderer has only four verb forms by default, so a goal whose only required object item is a location, or whose items are all unsure, renders no command; on the egocentric clip Sol’s one requirement was unsure. Where commands rendered they were mostly plausible (“Put the slice of bread in the white toaster then press the toaster lever”, Opus; “Put the eye drops box in the basket”, Gemini), and some carried visible errors: three outputs called the drawing paper a “tablet”, and the three strokes became “a cross” or “a small hash mark”.

The gripper signal as an optional source. With the signal as the event source on the SO-101 clip, Gemini kept the same phase sequence and the same three attempts (two failed) as its video-only run, and its coarse onsets agreed with the video-only coarse onsets at F1 0.71 within 10 frames (0.43 after the crawl). Sol also kept three attempts (F1 0.91 coarse, 0.73 after the crawl). Luna changed most: 13 onsets against 8, 16 segments against 12, and six attempts with five failed against three with two failed. The rater scored Luna’s output with the signal 2 and without it 4, Sol’s 5 with the signal and 2 without, and Gemini’s 5 both ways. The same signal thus moved the labels of different models in different directions.

6.3 E3: generality

No forcing on the dance, and no invented end state. On the dance clip, all six models returned a single segment of class `other` for the whole clip, set `has_end_state` to false, wrote no object end state, and rendered no command: every E3 end-state check passed. The rater nevertheless judged all six dance outputs not usable (scores 1 to 3). One segment for 18.9 s of dance is not forced into pick and place, and it also says little about the dance.

Partial forcing on the other human and tool clips. On the three clips where pick and place fits only partly, the share of segments with a pick-and-place class ranged from 0.15 (Luna, 3 of 20) through 0.22 (Sol, 7 of 32; Opus, 4 of 18) and 0.27 (Muse, 4 of 15) to 0.42 (Astra, 21 of 50) and 0.47 (Gemini, 16 of 34). The share

Table 11: Ratings per model, one rater, overall score from 1 to 5. “Blind only” leaves out the six cards whose model the rater had been told before saving the ratings (Section 6.4); n in brackets. The earlier round rated the V-lite pipeline with the gripper signal as boundary candidates, on four episodes. “Blend” is the rater’s own 70/30 ranking from the project README, a qualitative summary rather than a statistic.

Model	v1.1, video only, six clips					Earlier round		
	Mean	Blind only	Usable	Fit (y/p/n)	Best	Mean (4)	Blend	\$ / clip
GPT-6 Astra	3.00	3.00 (6)	3 of 6	3/3/0	1	3.25	3.08	0.522
Muse Spark 1.3	3.17	3.20 (5)	4 of 6	2/4/0	0	2.75	3.04	0.091
Claude Opus 5.5	3.00	2.60 (5)	2 of 6	2/4/0	2	3.00	3.00	0.149
GPT-6 Sol	2.83	1.75 (4)	2 of 6	2/3/1	2	3.25	2.96	0.094
Gemini 3.8 Flash	2.83	2.25 (4)	2 of 6	2/4/0	1	2.75	2.81	0.154
GPT-6 Luna	2.17	2.17 (6)	2 of 6	1/3/2	0	3.00	2.42	0.006

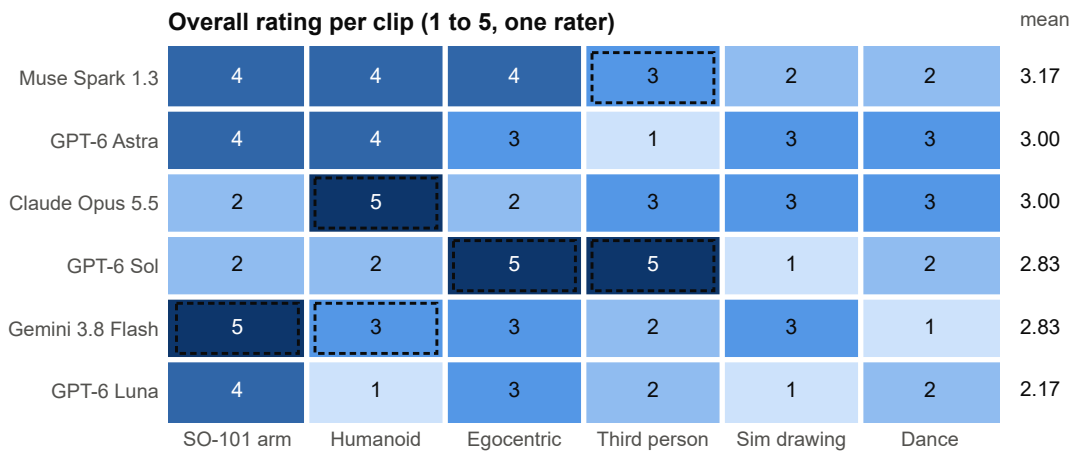


Figure 7: Overall rating of every main-arm output, one rater, on a blind page. Dashed outlines mark the six cards whose model the rater had been told before saving the ratings: four that the rater had asked to turn into video figures, having found them impressive, and two named by mistake while those figures were made (one of these two, the Opus humanoid card, later replaced the requested Gemini card in the README’s humanoid figure). Rows are ordered by the mean.

of phase texts that used a pick-and-place word was 0 for four models, 0.05 for Luna and 0.18 for Gemini. The larger class shares came from Gemini on the egocentric clip (11 of 20 segments, mostly approach) and Astra on the third-person clip (13 of 23). The open vocabulary was used: the outputs contain fold, wipe, press, pour and open classes, and `other` with a free-text description wherever no class fit.

6.4 Human ratings

Table 11 gives the per-model ratings and Figure 7 every main-arm rating. The mean overall score across the six models ranged from 2.17 (Luna) to 3.17 (Muse). No model produced labels rated usable on every clip: each had at least two clips rated not usable. No output for the dance or the drawing clip was rated usable. By clip, the SO-101 clip had the highest mean (3.50 over the six main arms), then the egocentric (3.33), humanoid (3.17) and third-person (2.67) clips, then the dance and the drawing (2.17 each). The rater’s best pick per clip went to Opus twice, Sol twice, and Gemini and Astra once each.

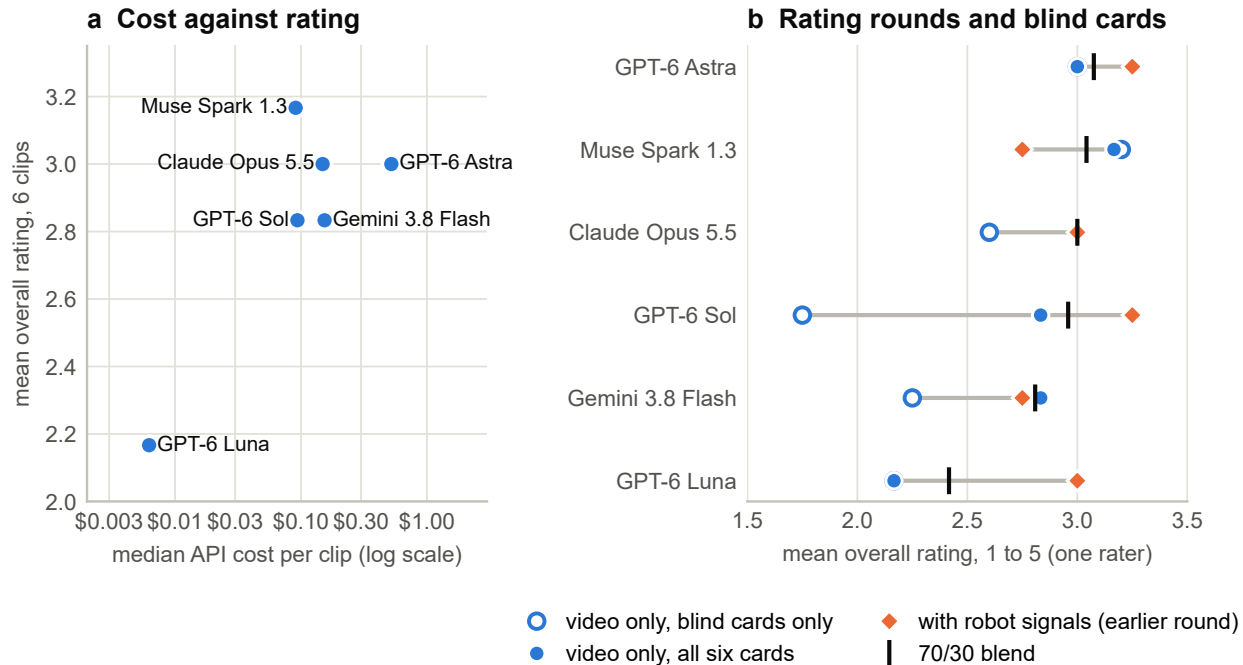


Figure 8: (a) Median API cost per clip against the mean overall rating of the v1.1 outputs. (b) Each model’s mean in the v1.1 round over all six cards (filled) and over its blind cards only (hollow), in the earlier round with robot signals (diamond), and the rater’s 70/30 blend of the two rounds (tick). One rater; six clips per model in v1.1 and four episodes in the earlier round.

Six cards were not blind. After the run, and before the ratings were saved, the rater asked for video figures of four cards they had found impressive, and the model behind each was named in the conversation. The rater’s page was an earlier build whose letters differed on two clips, so the first renders of two figures showed other cards, which were also named before being deleted. Six of the 36 main-arm cards were therefore not blind when the ratings were saved; whether any of them had been scored before they were named is not recorded. All four scores of 5 among the main-arm cards went to these cards (the round’s other two 5s went to the Gemini and Sol gripper extras), and so did four of the six best picks. Three of the four cards requested as figures scored 5, and all four had been judged impressive while still blind, so their high scores are partly a selection effect; the fourth, Gemini’s humanoid card, scored 3. The two named by mistake had no prior selection. One of them, Opus’s humanoid card, scored 5 and, after the ratings were saved, replaced the requested Gemini card in the README’s humanoid figure. Leaving out all six cards, Sol’s mean falls from 2.83 to 1.75, Gemini’s from 2.83 to 2.25 and Opus’s from 3.00 to 2.60; the rank correlation between the all-card and the blind-only means over the six models is 0.88.

The two rounds disagree. The earlier round’s per-model means (Table 11) rank the six models differently from the v1.1 round: the Spearman correlation between the rounds is -0.18 (Kendall -0.16), and -0.36 with the blind-only means. Luna fell most (3.00 to 2.17) and Muse rose most (2.75 to 3.17). The rounds differ in more than the signal: the pipeline version, the prompts, the episodes (three SO-101 and one LIBERO episode against six domains), the number of cards per page, and the rater’s familiarity with the task. The one episode rated in both rounds, the SO-101 clip, averaged 2.50 with the signal and 3.50 from video alone. Figure 8 shows both rounds, the blind-only means and the blend.

Price does not buy a better rating here. Across the six models, the median cost per clip and the mean rating have a Spearman correlation of 0.26 (0.38 with the mean cost). Muse at \$0.09 per clip had the highest mean, and Astra, at 3.5 times Opus’s cost, the same mean as Opus. With six clips per model, one rater and the confounds above, these are observations about this sample and do not rank the models.

Ratings judge the whole label. On the SO-101 clip, the rater’s scores and the refined boundary F1 at 5 frames were negatively rank-correlated across the six models (−0.62): Opus, with the highest F1 (0.40), and Sol (0.35) were rated 2, while Luna and Muse, with an F1 of 0, were rated 4. The overall score rates the phases, attempts, failures and goal together, and timing within 5 frames is only one part of a usable label.

6.5 Qualitative examples

Figure 1 shows one frame of each of the four video figures that the project README plays. Three are cards the rater picked as impressive, and the fourth, the humanoid panel, is the rater’s top-rated card on that clip; they show the pipeline at its best, and typical outputs are weaker (Section 6.4). On the SO-101 clip, where the gripper signal shows four missed grasps and an aborted close before the success, Gemini split the episode into three attempts of approach, grasp and retract: its first two grasps are marked failed (“Gripper attempts to grasp the eye drops box but slips and misses”), and the third is followed by transport, release and a final retract; the objective reads “The eye drops are inside the basket” and the command “Put the eye drops box in the basket”. On the humanoid clip, Opus marked a first grasp of the bread that fails to lift it, a back-off and a second, successful grasp, the bread carried and inserted into the toaster, and then the left hand pressing the lever; its command is “Put the slice of bread in the white toaster then press the toaster lever”. On the egocentric clip, Sol alternated fold and smoothing phases over 19 segments, including the shirt leaving the view and being unfolded and folded again; its one requirement (the shirt ends folded) was marked unsure, so no command rendered and the figure shows the objective. On the third-person clip, Sol labeled ten phases, from rinsing and squeezing the sponge and dispensing soap onto it to scrubbing and rinsing the bowl and then the plate, with classes wipe, pour and transport and free-text `other` phases.

6.6 Cost and operations

The v1.1 experiments sent 1,724 HTTP requests, each reserved against the budget before it went out, between 23:20 and 01:54 UTC, and were billed \$8.16 by the spend ledger (\$8.05 by the key’s own usage record; the \$0.11 difference was not investigated, and the ledger, the higher figure, is the one we report): \$1.58 for E1, \$6.55 for E2 and \$0.02 for debug calls. Astra alone was billed \$3.22. Fourteen requests were lost to dropped connections. The guard held each at its worst-case cost, \$1.02 in all, and the key’s usage later showed none of them had been billed. Six requests were refused by the guard, all in E2, and each refusal came in the same second as a lost request on the same model and clip, whose worst-case hold pushed that model’s budget bucket past its cap. Under the planned \$7.50 cap, 32 of the 41 E2 jobs had finished when the first pass ended; three had stopped at their budget buckets and six had not started. The author then added \$10 to the account and raised the run cap to \$8.50, keeping the \$1.50 balance floor, so that every model could run every clip; the second pass finished the remaining 9 jobs without a loss or a refusal. The ledger was compared with the key’s usage 176 times and never drifted past the limit. Luna met 32 rate-limit responses, all before any generation and billed at \$0. No held-out episode was accessed. With the V-lite sweep of September 27 (\$3.49), the V-lite and v1.1 experiments together cost about \$11.65 in API fees.

7 Discussion

Which onset should a label mark? The crawl’s two opposite results describe one behavior seen through two truths. Its refined onsets land a few frames after the measured start of finger motion, and the command starts 3 to 4 frames before that. A label meant for a vision-language model or a human reviewer arguably wants the moment the change becomes visible. A label that segments an action stream for a policy arguably wants the moment the command changes, since that is where the action data change. The data do not settle the choice, and the choice flips the sign of the result, so we report both and suggest that a labeled dataset state which onset its boundaries mark. The offset-corrected errors after the crawl (1.4 to 2.6 frames against the measured onset, 1.5 to 3.3 against the command) suggest that a constant per-dataset shift would remove between a third and four fifths of the remaining error, but that shift was fitted on the scored episodes and has to be tested on held-out ones before it can be used.

We can only offer hypotheses for why the crawl lands late. The question asks for the first frame where the fingers “have started to” close, and the first few frames of motion change a gripper by little at a long side of 448 pixels, so a conservative answer is the natural one. Stage 2 can also only move the onset between the image before the stage-1 pick and the pick: when stage 1 picks an image one step late, stage 2 cannot recover. Occlusion is a third factor and can push either way: in a debug filmstrip of a PickPlace episode, the arm hid the fingers at the brick, stage 1 picked a frame 12 frames before the truth, stage 2 then said the close had not begun by that frame, and the contradiction rule kept the stage-1 pick. A crawl that also asked for the last frame before the event, or that showed crops around the hand, would test these explanations.

Why one dataset and not the other. On `svla_sol01_pickplace`, the coarse boundaries sat a median of 7 frames early and the crawl had room to correct them. On `ArmNetBench`, the coarse boundaries were already centered on the measured onset (median signed error +1 frame, median absolute error 4 frames), at 20 frames per second each frame is 50 ms, and the crawl’s late bias shifted them to a median of +4. The `ArmNetBench` episodes are also harder in ways that the crawl cannot address: the policy failures hold most of the events, and the coarse passes proposed few of them. A refinement step helps most when the proposal is far off and the event is clearly visible, and it cannot help with an event that was never proposed.

Omission is the larger problem. In every configuration, about two thirds of the true gripper events had no prediction of the same type within a second, and the crawl, which moves boundaries and never adds one, left that unchanged. Part of this is the truth: an SO-101 gripper that opens during the approach or closes after the release produces an event that a phase label need not mark, so recall against gripper onsets has a ceiling below 1 for any correct phase labeler, and we did not measure how far below. Much of it is the coarse pass folding retries into one attempt: on the policy-failure episodes of `ArmNetBench`, Luna proposed 13 typed events against 214. Two directions follow. A recall pass could scan the clip with dense windows and ask only whether the fingers close or open there, and a free event source could propose windows for the crawl to check rather than candidates for the coarse pass to confirm. The motion candidates, given to the coarse pass as hints, made no clear difference; used as crawl windows they would be tested on recall directly.

Video as a provider samples it. Our video result concerns Gemini’s native video input at the provider’s default sampling, which the token counts suggest is about one frame per second. Frames at 2 per second carry more timing information, and the video proposals landed a pooled median of 12 frames early. The result does not show that video input localizes worse in general. A provider option for the sampling rate, where one exists, would allow a direct test at a matched rate.

Signals as optional evidence. V-lite showed what happens when models are given a signal’s candidates: they confirm almost all of them (252 of 270), their boundaries line up with the signal, the models look alike, and the pipeline fails silently where the signal has no event, as in the resets after a missed grasp. v1.1 reverses the default. The video decides and a signal is one more hint, so a labeler runs the same way on a humanoid, a person or a robot without a gripper channel. On the one clip where we ran both, the signal moved different models’ labels in different directions: Luna’s output got worse in the rater’s eyes, Sol’s got better, and Gemini’s stayed the same. By design, we did not score the gripper extras against gripper truth, so we cannot yet say whether the signal improves v1.1’s timing.

Choosing models. For the crawl, a cheap model looks sufficient against the measured onset. On the 15 episodes both ran, a Gemini crawl gave nearly the same F1 at 5 frames as the Luna crawl (descriptive, interval -0.076 to $+0.077$) at 27 times its cost; against the commanded onset it scored higher ($+0.148$, interval 0.083 to 0.210 , descriptive), so the choice interacts with the onset definition and has not been tested beyond these 15 episodes. For the full pipeline, cost per clip spanned a factor of 84 and did not track the rater’s scores. The ratings themselves are too thin to rank the models: one rater, six clips, two rounds that disagree, and six cards that were no longer blind. What they do show is that every model produced unusable labels on at least two of the six clips. robolabel routes outputs to review and reports a risk per episode, but in E2 the routing caught only 5 of the 21 main-arm outputs the rater judged unusable, so the checks alone do not yet find the unusable labels.

What the measurement machinery made possible. Hidden robot signals gave 420 measured and 443 commanded truth events on 60 episodes for no labeling effort. The response cache let the crawl contrasts share byte-identical coarse proposals, so they isolate the crawl. The spend guard let 1,724 requests run unattended under a hard cap. Its rule of holding a dropped request at its worst case is conservative by design: by our reading of the ledger, that rule caused all six refusals in this run, none of which was needed in hindsight, and we accept that cost in exchange for never overspending. Every v1.1 number in this paper traces to a receipt, a ledger line or a scored CSV file.

Running the experiments unattended. The v1.1 code and all three experiments were produced in one unattended session by an AI coding agent working from a written brief and specification, with the author available for decisions. The permission layer of the agent’s environment refused two actions until the author approved them explicitly: loosening the budget during the run, and later merging the work into the main branch. Two features of the record were fixed before any scored call: the pre-declared contrast list with its outcome words, and the frozen prompt state. The unblinding of six cards happened in the same session, when the author asked for figures of four cards and the agent named the model behind each, and behind two more by mistake. Keeping the rating page apart from every tool that knows the key would have prevented it.

8 Limitations

- **Development data only.** Every v1.1 result comes from development episodes and clips. E1 is exploratory by design: no held-out episode was used, and the five worded contrasts carry no correction for multiplicity.
- **Narrow timing truth.** Timing truth covers gripper onsets on two SO-101 datasets from one camera each. Contact boundaries were crawled but not scored in E1, and the gripper onset is a proxy that includes events a phase label need not mark. The offset-corrected MAE is in-sample.

- **One clip per domain.** E2 and E3 use one clip per domain, chosen for this run under the author’s brief; all their results are descriptive. The E2 truth on the humanoid clip covers the right hand only.
- **One rater.** All ratings come from one rater, who is also the system’s designer, in quick mode (about 19 minutes for the v1.1 round and 30 for the earlier one). Six of the 36 main-arm cards were no longer blind when the ratings were saved, and the two rating rounds differ in pipeline, prompts, episodes and rater familiarity.
- **A snapshot of models.** The models, versions and prices are those of one router on September 27 and 28, 2026. Router fallbacks were allowed, so the serving provider could vary. Reasoning ran at effort low only, and no model was sampled more than once per request, so run-to-run variance is unmeasured.
- **Rough timing.** Wall times are confounded by concurrency and rate limits in the first E2 pass.
- **Unknown video sampling.** The provider’s sampling rate for native video is unverified; robolabel neither sets nor records it.

9 Conclusion

robolabel drafts phases, attempts, failures and goals for robot, humanoid and human video, and measures how far those drafts can be trusted. Three generations of the pipeline showed that sparse frames cap timing, that anchoring boundaries to robot signals makes the labels look alike while hiding what the model contributes, and that a video-first design with a rule-triggered coarse-to-fine crawl moves proposed boundaries to a few frames after the measured onset, for about a third of a cent per episode. The same measurements show where the method stands: the crawl’s gain depends on which onset counts as truth and on the dataset, about two thirds of true gripper events have no prediction of the same type within a second, and no model we tried gives labels a rater would accept on every domain. The next steps follow from these results: decide and test a per-dataset onset shift on held-out episodes, add a recall pass for the events the coarse pass misses, and collect ratings from more than one rater with the blind page kept apart from every other tool.

Code and data

robolabel is open source under the Apache-2.0 license at <https://github.com/kevdozer1/robolabel>, with the v1.1 pipeline, the v8 prompts, the evaluation module and an offline demo. The blind rating site and the experiment scripts are not in the public repository. The experiments used these sources:

- PickPlace:
huggingface.co/datasets/lerobot/svla_sol101_pickplace
- ArmNetBench v0.1 (Apache-2.0):
huggingface.co/datasets/armnet/armnetbench_v01_lerobot_sol101
- Unitree G1 Dex3 toasted bread (Apache-2.0):
huggingface.co/datasets/unitreerobotics/G1_Dex3_ToastedBread_Dataset
- Eidon Tracker POV by Eidon AI (Solidic Labs Inc), recording 8266 (CC BY 4.0):
huggingface.co/datasets/eidon-ai/tracker-pov
- “Son del Zapateado” by Koffer and “Washing Dishes” by Leet289, both on Wikimedia Commons (CC BY-SA 4.0).

- The drawing clip, rendered from the author’s statebench project with robot models from MuJoCo Menagerie [32].

The per-episode scores, the ratings and the run receipts are not part of the public repository and are available from the author on request.

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